CHAPTER 6 WORK AND ENERGY

6.1 Meaning of work. In physics, work is done only when a force moves an object along a path. The formula is W = F x d, where W is work, F is the force in the direction of the motion, and d is the displacement. The SI unit of work is the joule J. One joule is the work done by a force of 1 newton acting through 1 metre. Worked example. A student pushes a crate with a force of 10 N along a smooth floor for 3 m. The force is in the same direction as the motion, so W = 10 x 3 = 30 J. If the crate does not move, the displacement is 0 m and the work is 0 J, even if the student feels tired.

6.2 DIRECTION OF THE FORCE

Only the part of the force along the displacement counts as work. If the force is at a right angle to the motion, that part is zero, so the work is zero. A student who carries a bag across a flat corridor at constant height does no work against gravity on the bag, because the weight is vertical and the path is horizontal. If the same student then lifts the bag onto a shelf, the force and the displacement point the same way, so work is done. Keep the units together: 30 J is not the same as 30 N.

6.3 KINETIC ENERGY

Energy is the capacity to do work. Kinetic energy is the energy of motion. The formula is KE = 1/2 m v^2, where m is mass in kilograms and v is speed in metres per second. Doubling the speed multiplies the kinetic energy by four, because the speed is squared. Worked example. A 2 kg trolley moves at 3 m/s. KE = 0.5 x 2 x 3 x 3 = 9 J. If the trolley slows to 1 m/s, KE becomes 1 J. The lost 8 J was transferred, often as heat in the brakes or as work done on another object.

6.4 POTENTIAL ENERGY

Gravitational potential energy depends on height. The formula is PE = m g h, where g is about 10 N/kg near Earth and h is the height gained. Worked example. A 2 kg mass is raised 5 m. PE = 2 x 10 x 5 = 100 J. That 100 J came from the work done in lifting. In a closed system the total energy stays the same: as a ball falls, potential energy becomes kinetic energy. A common mix-up is to treat height as speed. Height is a position; speed is how fast the position is changing.

6.5 POWER

Power is how quickly work is done. The formula is P = W / t, where t is time in seconds. The SI unit is the watt W. One watt means one joule each second. Worked example. A motor does 60 J of work in 2 s. P = 60 / 2 = 30 W. A second motor that does the same 60 J in 6 s has a power of only 10 W. Power does not tell you the total energy by itself; you also need the time. Cover the formula and say it: power equals work divided by time.

6.6 CHAPTER REVIEW

Hold these relations together. Work: W = F x d. Kinetic energy: KE = 1/2 m v^2. Potential energy: PE = m g h. Power: P = W / t. Units stay with the quantity: N with force, m with distance, J with work and energy, W with power. Practice A. A 8 N force pushes a box 5 m in the same direction. What is the work? Practice B. That work is done in 4 s. What is the power? Practice C. Why is the answer to A written in joules, not in newtons? Check yourself by covering this page and writing the four formulas from memory.